

EXPERIMENT 1

Traditional Development vs Low-Code Development - Comparative Case Study

AIM

To examine and compare traditional software development with low-code development with respect to coding effort, development duration, flexibility, cost, customization, and ease of implementation through a practical case study.

OBJECTIVE

To identify the major differences between traditional and low-code approaches and understand which approach is appropriate for different types of software applications.

INTRODUCTION

Software applications may be built using conventional programming methods or modern low-code platforms. Traditional development involves writing application logic manually with programming languages, frameworks, and database technologies. Low-code development uses visual designers, reusable components, drag-and-drop controls, and automated workflows to reduce the amount of hand-written code.

CASE STUDY

A Student Management System is considered to compare both approaches. The system contains student registration, record management, authentication, database operations, and a user interface.

Traditional Development

In the traditional approach, developers build the user interface, server-side logic, database connectivity, authentication, and other modules by writing code. This approach offers detailed control and supports extensive customization, but it normally requires more development effort and programming expertise.

Low-Code Development

With a low-code platform, developers assemble the application using visual development features, ready-made components, connectors, and workflow automation. This reduces manual coding and can shorten the time needed to create and modify the application.

COMPARISON

Feature	Traditional Development	Low-Code Development
Coding Effort	High	Low
Development Duration	Usually longer	Usually shorter
Flexibility	Very high	Moderate
Technical Expertise	Strong programming skills	Less manual coding required
Customization	Extensive customization	Depends on platform features
Cost	Can be higher	Can be lower
Maintenance	Developer involvement needed	Often simpler for standard applications

OBSERVATION

The comparison shows that traditional development provides greater control, flexibility, and scope for custom features. However, it generally needs more programming effort and a longer development cycle.

Low-code development minimizes repetitive coding through visual tools and reusable modules. It can support faster delivery, particularly for straightforward business applications and systems with common workflows.

RESULT

The two software development approaches were compared successfully using the Student Management System case study. Traditional development is appropriate when extensive customization and detailed control are required, while low-code development can be used when rapid implementation and reduced manual coding are priorities.